Electrodynamics
Independence Semester 2024

Homework 1

August 18, 2024

Problem 1

Maxwell's equations in free space are given in SI units:

$$\vec{\nabla} \cdot \vec{E}' = \frac{\rho'}{\epsilon_0} \, , \, \vec{\nabla} \times \vec{B}' - \mu_0 \epsilon_0 \frac{\partial \vec{E}'}{\partial t} = \mu_0 \vec{J}' \, , \, \vec{\nabla} \cdot \vec{B}' = 0 \, , \, \vec{\nabla} \times \vec{E}' + \frac{\partial \vec{B}'}{\partial t} = 0. \tag{1}$$

Maxwell's equations in free space are given in Gaussian units:

$$\vec{\nabla} \cdot \vec{E} = 4\pi\rho \, , \, \vec{\nabla} \times \vec{B} - \frac{1}{c}\frac{\partial \vec{E}}{\partial t} = \frac{4\pi}{c}\vec{J} \, , \, \vec{\nabla} \cdot \vec{B} = 0 \, , \, \vec{\nabla} \times \vec{E} + \frac{1}{c}\frac{\partial \vec{B}}{\partial t} = 0. \tag{2}$$

Please assume linear relationships between primed and unprimed quantities, i.e.,

$$\vec{E}' = \alpha \vec{E} \, , \, \vec{B}' = \beta \vec{B} \, , \, \rho' = \gamma \rho \, , \, \vec{J}' = \delta \vec{J}. \tag{3}$$

(a) Evaluate expressions for α, β, γ, δ using the two sets of Maxwell's equations given above.

(b) Write down the primed quantities, i.e. $\vec{E}'$, $\vec{B}'$, ρ', $\vec{J}'$ in terms of the corresponding unprimed quantities.

Hint: You will require to use the Coulomb's law to obtain the '4π' factors in α, β, γ, δ.

Problem 2

Assume static $\vec{E}$ and $\vec{B}$ fields. Please recall that you will get the same $\vec{E}$ and $\vec{B}$ from the scalar (ϕ) and vector ($\vec{A}$) if they transform as:

$$\phi \to \phi' = \phi - \frac{1}{c}\frac{\partial \lambda}{\partial t} \, , \, \vec{A} \to \vec{A}' = \vec{A} + \vec{\nabla}\lambda. \tag{4}$$

The above transformations are known as gauge transformations.

(a) If $\vec{E}$ and $\vec{B}$ are static, i.e., time-independent, it is convenient to assume that ϕ and $\vec{A}$ are also time-independent. If ϕ and $\vec{A}$ are time-independent, write down the most general function for $\lambda(t, \vec{r})$ that will achieve that purpose.

(b) Using the most general function for $\lambda(t, \vec{r})$ of part (a), write down the gauge transformations of static $\vec{E}$ and $\vec{B}$ fields.

Problem 3

Prove explicitly that $\nabla^2 \left(\frac{1}{|\vec{r}-\vec{r'}|} \right) = -4\pi\delta^3(\vec{r} - \vec{r'})$.

Problem 4

Consider a system where a conducting surface is placed in free space. Please recall the boundary conditions at the interface between two media for $\vec{E}$ and $\vec{D}$, that is

$$\hat{n} \cdot (\vec{D}_2 - \vec{D}_1) = 4\pi\sigma \, , \; \hat{n} \times (\vec{E}_2 - \vec{E}_1) = 0. \tag{5}$$

(a) Assuming $\vec{E}$ only on one side of the conductor, derive the expression for the surface charge density 'σ'.

(b) Derive the expression for 'σ' if there are electric fields on both sides of the conductor.

(c) Express the pressure on the conducting surface in terms of 'σ' for case (a).

(d) Can we express the pressure on the conducting surface in terms of 'σ' for case (b)? If not, why?

Problem 5

(a) Writing $\vec{E}$ and $\vec{B}$ in terms of gauge potentials ϕ and $\vec{A}$, obtain the equations for ϕ and $\vec{A}$ that result from plugging $\vec{E}$ and $\vec{B}$ into the two Maxwell's field equations (in Gaussian unit system). Assume there is no medium present, so $\vec{D} = \vec{E}$ and $\vec{H} = \vec{B}$.

(b) It is possible to choose a gauge where $\nabla \cdot \vec{A} + \frac{1}{c}\frac{\partial \phi}{\partial t} = 0$ (this is known as Lorenz gauge). Show that in Lorenz gauge, the equations for ϕ and $\vec{A}$ that you obtained in part (a) become simply wave operators acting on ϕ and $\vec{A}$, with ρ and $\vec{J}$ acting as source terms.

Problem 6

The time-averaged potential of a neutral hydrogen atom is given by

$$\phi = \frac{qe^{-\alpha r}}{r}\left(1 + \frac{\alpha r}{2}\right), \tag{6}$$

where q and α are constants. Find the distribution of charge both continuous (and discrete) that will give this potential, interpret your result physically.

Problem 7

a) Using the convention that any two-index Cartesian 3-tensor T_{ij} can be viewed as a matrix T with rows labelled by i and columns labelled by j, rewrite the following equations in matrix notation:

$$D_{ij} = A_{ik}B_{kl}C_{jl}, \quad D_{ij} = A_{ik}B_{jl}C_{kl}, \quad D_{ij} = A_{ik}(B_{kj} + C_{jk}).$$

b) Show that if W is any 3×3 matrix, with components W_{ij}, then $W_{il}W_{jm}W_{kn}\epsilon_{lmn} = (\det.W)\,\epsilon_{ijk}$. (Hint: First show that the left-hand side is totally antisymmetric in ijk, and hence that it must be proportional to ϵ_{ijk}. This reduces the labour greatly, since then there is only one non-trivial case to be checked.)

Problem 8

a) Consider a Lorentz boost with velocity $\vec{v} = (v, 0, 0)$ along the x-axis followed by a Lorentz boost with velocity $\vec{\tilde{v}} = (0, \tilde{v}, 0)$ along the y-axis. Thus the first boost transforms from (x, y, z, t) to (x', y', z', t') and the second transforms from (x', y', z', t') to (x'', y'', z'', t''), where

$$x' = \gamma(x - vt)\,,\ y' = y\,,\ z' = z\,,\ t' = \gamma(t - vx)\,, \tag{7}$$

$$x'' = x'\,,\ y'' = \tilde{\gamma}(y' - \tilde{v}t')\,,\ z'' = z'\,,\ t'' = \tilde{\gamma}(t' - \tilde{v}y')\,, \tag{8}$$

with $\gamma = (1 - v^2)^{-1/2}$ and $\tilde{\gamma} = (1 - \tilde{v}^2)^{-1/2}$. Show that the final result of the two successive boosts can be expressed as the composition of a single Lorentz boost from (x, y, z, t) to $(\hat{x}, \hat{y}, \hat{z}, \hat{t})$, with

$$\vec{\hat{r}} = \vec{r} + \frac{\hat{\gamma} - 1}{\hat{v}^2}(\vec{\hat{v}} \cdot \vec{r})\vec{\hat{v}} - \tilde{\gamma}\vec{\hat{v}}t\,,\ \hat{t} = \hat{\gamma}(t - \vec{\hat{v}}.\vec{r})\,,\ \text{with}\ \hat{\gamma} \equiv (1 - \hat{v}^2)^{-1/2}\,, \tag{9}$$

for a certain velocity vector $\vec{\hat{v}}$, followed by a pure spatial rotation, giving the final result

$$\begin{pmatrix} x'' \\ y'' \\ z'' \end{pmatrix} = \begin{pmatrix} \cos\psi & -\sin\psi & 0 \\ \sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \hat{x} \\ \hat{y} \\ \hat{z} \end{pmatrix}\,,\ t'' = \hat{t}\,,$$

where (x'', y'', z'', t'') are given in Eq. (8). You should calculate the velocity vector $\vec{\hat{v}}$ and $\tan\psi$ in terms of original pair of velocities v and $\tilde{v}$.

b) Show that there is a very simple expression for $\hat{\gamma}$ in terms of γ and $\tilde{\gamma}$.

c) It is sometimes useful to define a quantity called the rapidity δ, related to the velocity by $v = \tanh\delta$. Show also that the rotation angle ψ has a very simple expression in terms of rapidities δ and $\tilde{\delta}$ ($v = \tanh\delta$, $\tilde{v} = \tanh\tilde{\delta}$) of the orthogonal boost velocities, when written using half-angles, which is given by

$$\tan\frac{\psi}{2} = \tanh\frac{\delta}{2}\,\tanh\frac{\tilde{\delta}}{2}\,. \tag{10}$$

This result, in which the composition of two non-parallel Lorentz boosts gives rise to a spatial rotation in addition to a boost, is a special case of what is known as the *Thomas-Wigner rotation*, which gives a relativistic correction to the precession of a gyroscope or a spinning particle.